Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2021

Bachelor in Computer Applications

Course Title: **Mathematics I**

Code No: Semester: **I**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. In an examination, 58% failed in Account, 39% in English, and 25% in Statistics; 32% in Accounts and English, 19% in Account and Statistics, 17% in English and Statistics, and 13% in all three subjects. Find:i) What percent passed in all three subjects?ii) What percent failed in exactly two subjects?

3. If $\sqrt{x + ty} = a + ib$, then find $\sqrt{x - ty}$ and $x^2 + y^2$.

4. A class consists of boys whose ages are in A.P. (common difference = 4 months). The youngest boy is 8 years old, and the total age of the class is 168 years. Find the number of boys.

5. If $A = \begin{pmatrix} 4 & 0 \\ 0 & 5 \end{pmatrix}$, find a matrix X such that $AX = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

6. Find the equation of the ellipse whose latus rectum is 3 and eccentricity is $\frac{1}{\sqrt{2}}$.

7. Prove by vector method: $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

8. A committee of 5 is to be constituted from 6 boys and 5 girls. How many ways can this be done to include at least one boy and one girl?

Group C

Attempt any TWO questions[2x10=20]

9. Define a function, its domain, and range. Find the domain and range of $f(x) = \sqrt{2 - x - x^2}$.

10. a) Three numbers in A.P. sum to 15. If 1, 3, 9 are added to them respectively, they form a G.P. Find the original numbers.b) Find the sum to n terms of the series $\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \cdots$.

11. a) Find the angle between vectors $\mathbf{u} = 4\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$.b) Find the Maclaurin series of $f(x) = \cos x$.